

Best Practices for Building Data Warehouses

1. Title Slide

- Topic: Best Practices for Building Data Warehouses
 - Subtitle: Strategy, Tools, Pitfalls, and Future Trends
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2. Recipe for a Successful Data Warehouse

Description:

- Understand business needs (data modeling, KPIs, reporting).
- Design scalable and flexible architecture.
- Choose the right ETL tools and integrate reliable metadata.
- Maintain data quality and governance practices.
- Ensure end-user accessibility with self-service BI tools.

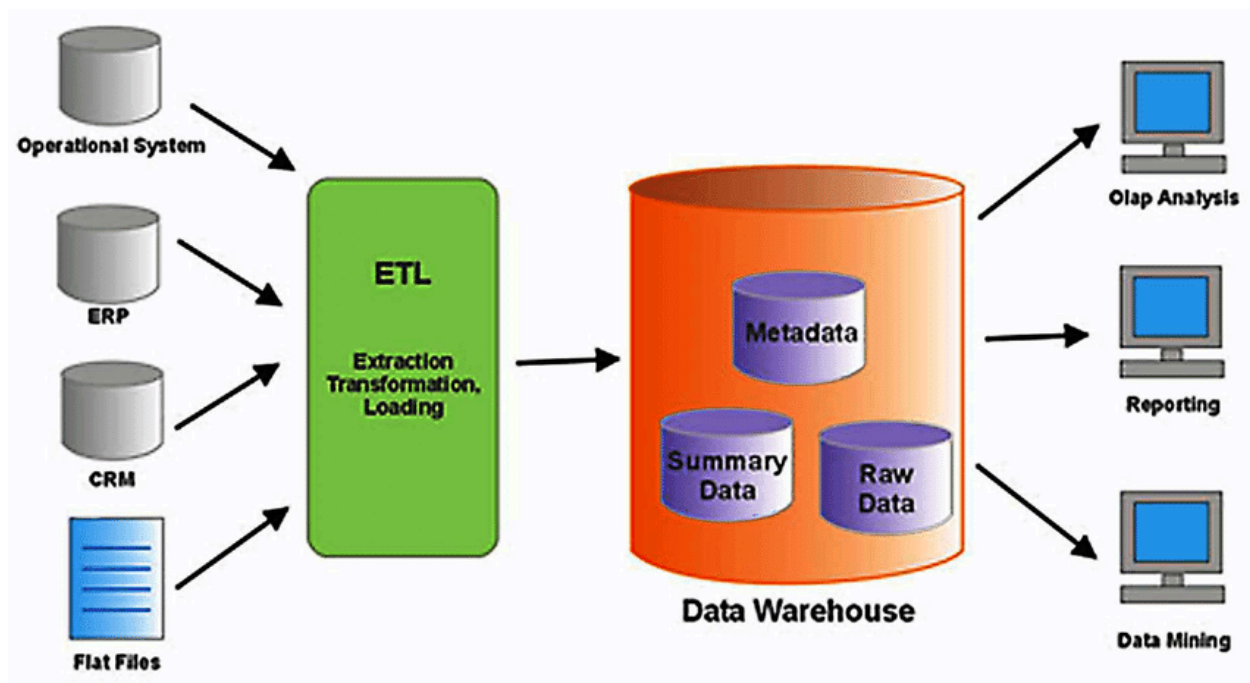


Illustration:

- Diagram: Successful DW Life Cycle (Analysis → Design → ETL → QA → BI Access)

Code Sample (ETL Tool Use):

```
-- Sample SQL to perform daily data load with deduplication
INSERT INTO warehouse.sales_summary (sale_date, product_id,
total_sales)
SELECT sale_date, product_id, SUM(sale_amount)
FROM staging.sales_data
WHERE sale_date = CURRENT_DATE - INTERVAL '1 day'
GROUP BY sale_date, product_id;
```

3. Common Pitfalls in Data Warehousing

Description:

- Lack of stakeholder involvement.
- Ignoring data quality and consistency.
- Underestimating ETL complexity.
- Not aligning DW design with business goals.
- Failing to adapt to change (agility issues).

Common Pitfalls in Data Analysis for Decision Making



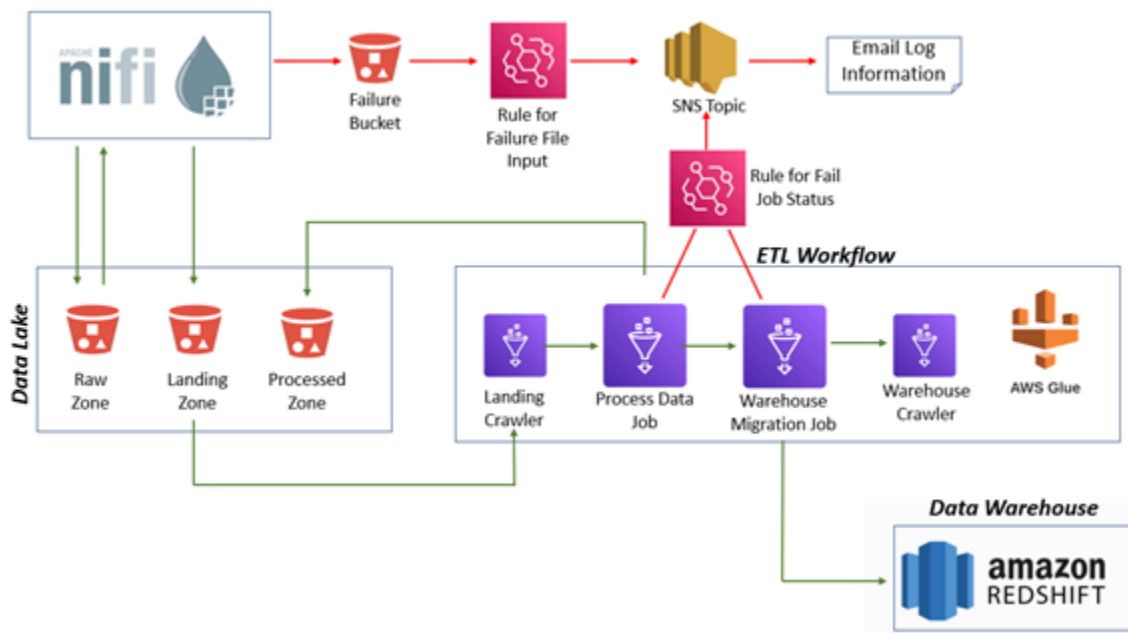
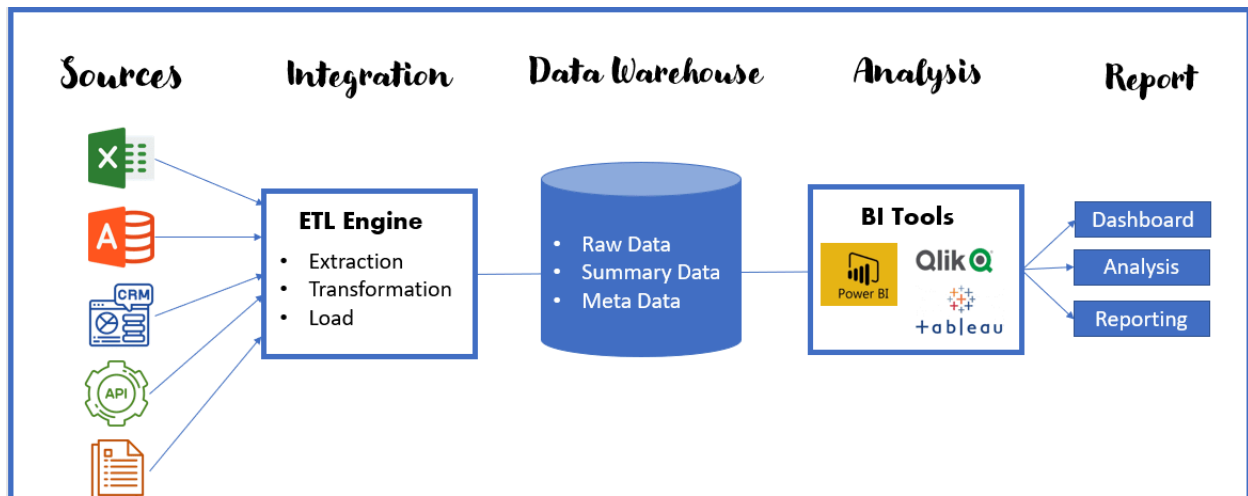
Illustration:

- Table: Pitfall vs Solution (e.g., “No business alignment” → “Involve stakeholders early”)
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4. Popular BI & DW Tools and Suites

Categories:

- **ETL Tools:** Talend, Informatica, Apache NiFi, AWS Glue
- **BI Tools:** Power BI, Tableau, Looker, QlikView
- **DW Platforms:** Amazon Redshift, Snowflake, Google BigQuery, Azure Synapse



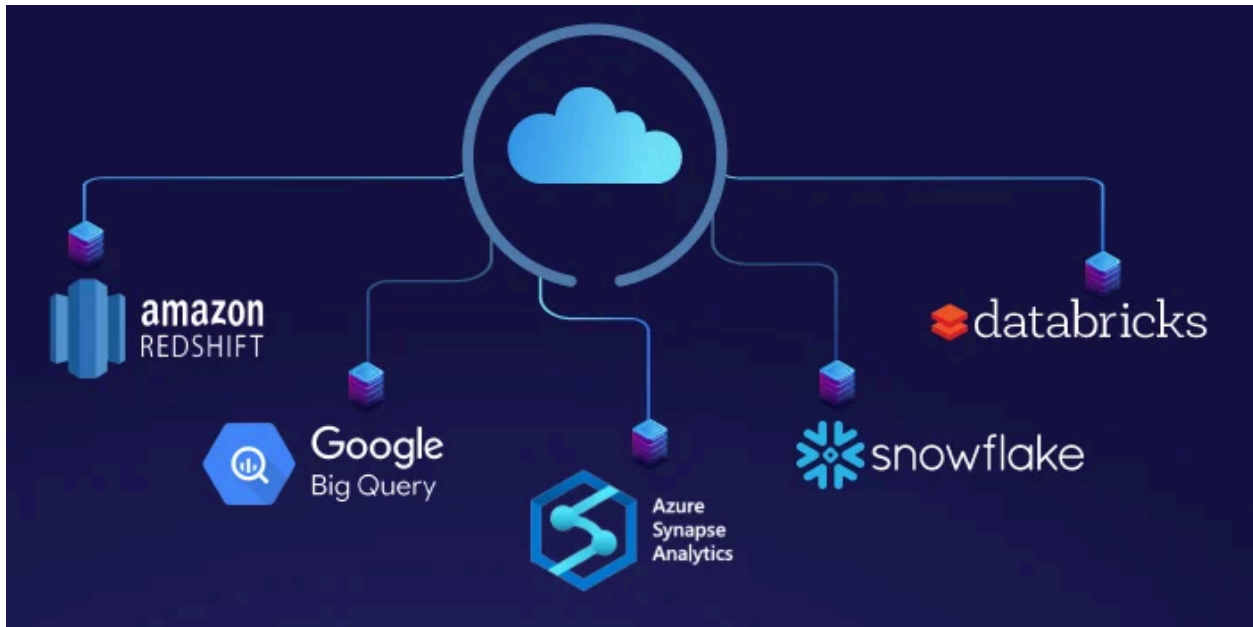


Illustration:

- Infographic or logo cloud of modern BI/DW tools

Code Sample (Power BI DAX Expression):

```
TotalSales = SUM(Sales[Amount])
```

5. Trends in BI/DW

Description:

- Shift to cloud-based data warehousing
- Rise of **DataOps** and **CI/CD for Data Pipelines**
- Adoption of **Data Lakehouse** architecture
- Integration of **AI/ML for predictive analytics**
- Real-time analytics and streaming ingestion (Kafka, Flink)



Illustration:

- Diagram: Modern Cloud-Native DW Stack

Code Sample (Streaming):

```
-- Example Kafka consumer reading real-time sales  
kafka-console-consumer.sh --topic real-time-sales --bootstrap-server  
localhost:9092
```

6. Conclusion

- Focus on business alignment
 - Use modern tools and best practices
 - Avoid known pitfalls
 - Be agile and adaptive to trends
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